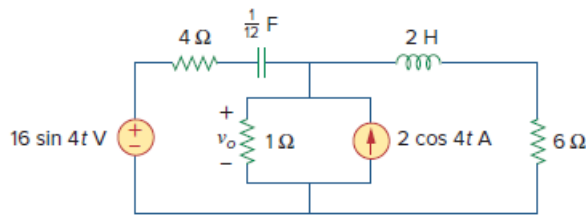


Problem

Determine v_o in the circuit of Fig. 10.52.

Figure 10.52:



Step-by-step solution

Step 1 of 4

Convert the circuit to the frequency domain.

$$16 \sin 4t = 16 \cos(4t - 90^\circ)$$

$$16 \cos(4t - 90^\circ) \text{ V} \Rightarrow 16 \angle -90^\circ \text{ V} \quad \omega = 4 \text{ rad/s}$$

$$2 \cos(4t) \text{ A} \Rightarrow 2 \angle 0^\circ \text{ A} \quad \omega = 4 \text{ rad/s}$$

$$\frac{1}{12} \text{ F} \Rightarrow \frac{1}{j\left(\frac{1}{12}\right)4} = -j3 \Omega$$

$$2 \text{ H} \Rightarrow j4(2) = j8 \Omega$$

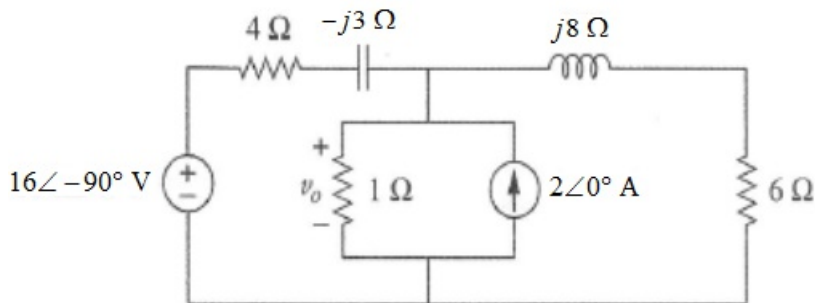


Figure 1

v_o

Step 2 of 4

Thus, the frequency domain equivalent circuit is shown in Figure 1.

$$\frac{v_o - 16 \angle -90^\circ}{4 - 3j} + \frac{v_o}{1} + \frac{v_o}{6 + 8j} = 2 \angle 0^\circ \text{ A}$$

$$v_o \left(\frac{1}{4 - 3j} + 1 + \frac{1}{6 + 8j} \right) = 2 \angle 0^\circ + \frac{16 \angle -90^\circ}{4 - 3j}$$

$$v_o \left(\frac{1}{4 - 3j} + 1 + \frac{1}{6 + 8j} \right) = \frac{2 \angle 0^\circ (4 - 3j) + 16 \angle -90^\circ}{4 - 3j}$$

$$v_o \left(\frac{6 + 8j + (4 - 3j)(6 + 8j) + 4 - 3j}{(4 - 3j)(6 + 8j)} \right) = \frac{2(4 - 3j) - 16j}{4 - 3j}$$

Step 3 of 4

Applying Kirchhoff's current law at node

$$v_o \left(\frac{6+8j+24+32j-18j+24+4-3j}{6+8j} \right) = 8-6j-16j$$

$$v_o \left(\frac{58+19j}{6+8j} \right) = 8-22j$$

$$v_o = \frac{(8-22j)(6+8j)}{(58+19j)}$$

$$v_o = \frac{48+64j-132j+176}{(58+19j)}$$

$$v_o = \frac{224-68j}{58+19j}$$

$$v_o = 3.140 - 2.2013j \text{ V}$$

$$v_o = 3.835 \angle -35.024^\circ \text{ V} \quad \omega = 4 \text{ rad/sec}$$

$$v_o(t)$$

$$\boxed{3.835 \cos(4t - 35.024^\circ) \text{ V}}$$

Step 4 of 4

For any sinusoidal input the output is also a sinusoid with change in magnitude and phase.

Hence the output voltage v_o is,

$$v_o = 3.835 \cos(4t - 35.02^\circ) \text{ V}$$

Thus, the voltage v_o is $\boxed{3.835 \cos(4t - 35.02^\circ) \text{ V}}$.